

The CYK Algorithm

- *The membership problem:*
 - Problem:
 - Given a context-free grammar **G** and a string **w**
 - **G** = (V, Σ, P, S) where
 - » V finite set of variables
 - » Σ (the alphabet) finite set of terminal symbols
 - » P finite set of rules
 - » S start symbol (distinguished element of V)
 - » V and Σ are assumed to be disjoint
 - **G** is used to generate the string of a language
 - Question:
 - Is **w** in **L(G)**?

The CYK Algorithm

- J. Cocke
 - D. Younger,
 - T. Kasami
- Independently developed an algorithm to answer this question.

The CYK Algorithm Basics

- The Structure of the rules in a Chomsky Normal Form grammar
- Uses a “dynamic programming” or “table-filling algorithm”

Chomsky Normal Form

- *Normal Form* is described by a set of conditions that each rule in the grammar must satisfy
- Context-free grammar is in CNF if each rule has one of the following forms:
 - $A \rightarrow BC$ at most 2 symbols on right side
 - $A \rightarrow a$, or terminal symbol
 - $S \rightarrow \lambda$ null stringwhere $B, C \in V - \{S\}$

Construct a Triangular Table

- Each row corresponds to one length of substrings
 - Bottom Row – Strings of length 1
 - Second from Bottom Row – Strings of length 2
 - .
 - .
 - Top Row – string 'w'

Construct a Triangular Table

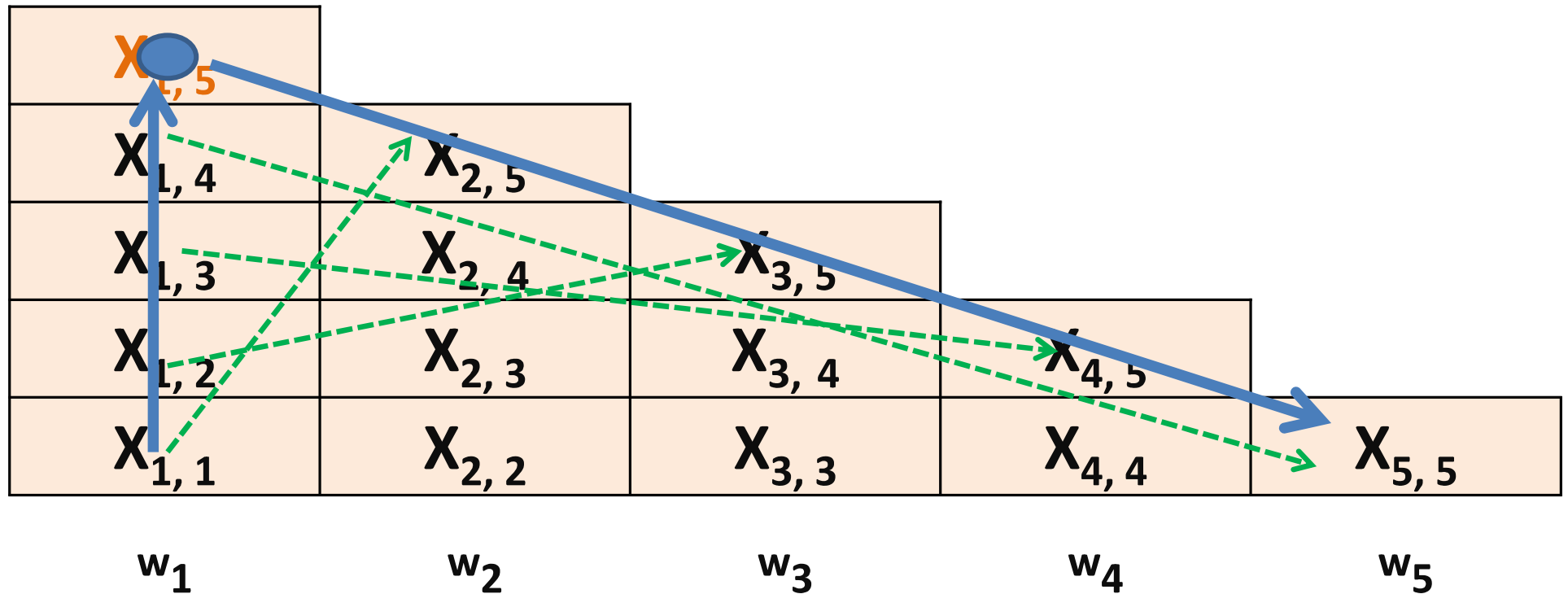
- $X_{i,i}$ is the set of variables A such that $A \rightarrow w_i$ is a production of G
- Compare at most n pairs of previously computed sets:
 $(X_{i,i}, X_{i+1,j}), (X_{i,i+1}, X_{i+2,j}) \dots (X_{i,j-1}, X_{j,j})$

Construct a Triangular Table

$X_{1,5}$				
$X_{1,4}$	$X_{2,5}$			
$X_{1,3}$	$X_{2,4}$	$X_{3,5}$		
$X_{1,2}$	$X_{2,3}$	$X_{3,4}$	$X_{4,5}$	
$X_{1,1}$	$X_{2,2}$	$X_{3,3}$	$X_{4,4}$	$X_{5,5}$
w_1	w_2	w_3	w_4	w_5

Table for string ' w ' that has length 5

Construct a Triangular Table



Looking for pairs to compare

Example CYK Algorithm

- Show the CYK Algorithm with the following example:
 - CNF grammar **G**
 - $S \rightarrow AB \mid BC$
 - $A \rightarrow BA \mid a$
 - $B \rightarrow CC \mid b$
 - $C \rightarrow AB \mid a$
 - **w** is baaba
 - Question Is **baaba** in $L(G)$?

Constructing The Triangular Table

{B}	{A, C}	{A, C}	{B}	{A, C}
b	a	a	b	a

$S \rightarrow AB \mid BC$

$A \rightarrow BA \mid a$

$B \rightarrow CC \mid b$

$C \rightarrow AB \mid a$

Calculating the Bottom ROW

Constructing The Triangular Table

- $X_{1,2} = (X_{i,i}, X_{i+1,j}) = (X_{1,1}, X_{2,2})$
- $\rightarrow \{B\}\{A,C\} = \{BA, BC\}$
- Steps:
 - Look for production rules to generate BA or BC
 - There are two: S and A
 - $X_{1,2} = \{S, A\}$

S $\rightarrow$ AB | BC

A $\rightarrow$ BA | a

B $\rightarrow$ CC | b

C $\rightarrow$ AB | a

Constructing The Triangular Table

{S, A}				
{B}	{A, C}	{A, C}	{B}	{A, C}
b	a	a	b	a

Constructing The Triangular Table

- $X_{2,3} = (X_{i,i}, X_{i+1,j}) = (X_{2,2}, X_{3,3})$
- $\rightarrow \{A, C\}\{A, C\} = \{AA, AC, CA, CC\} = Y$
- Steps:
 - Look for production rules to generate Y
 - There is one: B
 - $X_{2,3} = \{B\}$

$S \rightarrow AB \mid BC$
 $A \rightarrow BA \mid a$
 $B \rightarrow CC \mid b$
 $C \rightarrow AB \mid a$

Constructing The Triangular Table

{S, A}	{B}			
{B}	{A, C}	{A, C}	{B}	{A, C}
b	a	a	b	a

Constructing The Triangular Table

- $X_{3,4} = (X_{i,i}, X_{i+1,j}) = (X_{3,3}, X_{4,4})$
- $\rightarrow \{A, C\}\{B\} = \{AB, CB\} = Y$
- Steps:
 - Look for production rules to generate Y
 - There are two: S and C
 - $X_{3,4} = \{S, C\}$

$S \rightarrow AB \mid BC$
 $A \rightarrow BA \mid a$
 $B \rightarrow CC \mid b$
 $C \rightarrow AB \mid a$

Constructing The Triangular Table

{S, A}	{B}	{S, C}		
{B}	{A, C}	{A, C}	{B}	{A, C}
b	a	a	b	a

Constructing The Triangular Table

- $X_{4,5} = (X_{i,i}, X_{i+1,j}) = (X_{4,4}, X_{5,5})$
- $\rightarrow \{B\}\{A, C\} = \{BA, BC\} = Y$
- Steps:
 - Look for production rules to generate Y
 - There are two: S and A
 - $X_{4,5} = \{S, A\}$

$S \rightarrow AB \mid BC$
 $A \rightarrow BA \mid a$
 $B \rightarrow CC \mid b$
 $C \rightarrow AB \mid a$

Constructing The Triangular Table

{S, A}	{B}	{S, C}	{S, A}	
{B}	{A, C}	{A, C}	{B}	{A, C}
b	a	a	b	a

Constructing The Triangular Table

- $X_{1,3} = (X_{i,i}, X_{i+1,j}) (X_{i,i+1}, X_{i+2,j})$
 $= (X_{1,1}, X_{2,3}), (X_{1,2}, X_{3,3})$
- $\rightarrow \{B\}\{B\} \cup \{S, A\}\{A, C\} = \{BB, SA, SC, AA, AC\} = Y$
- Steps:
 - Look for production rules to generate Y
 - There are NONE: S and A
 - $X_{1,3} = \emptyset$
 - no elements in this set (empty set)

$S \rightarrow AB \mid BC$
 $A \rightarrow BA \mid a$
 $B \rightarrow CC \mid b$
 $C \rightarrow AB \mid a$

Constructing The Triangular Table

$\emptyset$				
$\{S, A\}$	$\{B\}$	$\{S, C\}$	$\{S, A\}$	
$\{B\}$	$\{A, C\}$	$\{A, C\}$	$\{B\}$	$\{A, C\}$
b	a	a	b	a

Constructing The Triangular Table

- $X_{2,4} = (X_{i,i}, X_{i+1,j}) (X_{i,i+1}, X_{i+2,j})$
 $= (X_{2,2}, X_{3,4}), (X_{2,3}, X_{4,4})$
- $\rightarrow \{A, C\}\{S, C\} \cup \{B\}\{B\} = \{AS, AC, CS, CC, BB\} = Y$
- Steps:
 - Look for production rules to generate Y
 - There is one: B
 - $X_{2,4} = \{B\}$

$S \rightarrow AB \mid BC$
 $A \rightarrow BA \mid a$
 $B \rightarrow CC \mid b$
 $C \rightarrow AB \mid a$

Constructing The Triangular Table

$\emptyset$	$\{B\}$			
$\{S, A\}$	$\{B\}$	$\{S, C\}$	$\{S, A\}$	
$\{B\}$	$\{A, C\}$	$\{A, C\}$	$\{B\}$	$\{A, C\}$
b	a	a	b	a

Constructing The Triangular Table

- $X_{3,5} = (X_{i,i}, X_{i+1,j}) (X_{i,i+1}, X_{i+2,j})$
 $= (X_{3,3}, X_{4,5}), (X_{3,4}, X_{5,5})$
- $\rightarrow \{A,C\}\{S,A\} \cup \{S,C\}\{A,C\}$
 $= \{AS, AA, CS, CA, SA, SC, CA, CC\} = Y$
- Steps:
 - Look for production rules to generate Y
 - There is one: B
 - $X_{3,5} = \{B\}$

$S \rightarrow AB \mid BC$
 $A \rightarrow BA \mid a$
 $B \rightarrow CC \mid b$
 $C \rightarrow AB \mid a$

Constructing The Triangular Table

$\emptyset$	$\{B\}$	$\{B\}$		
$\{S, A\}$	$\{B\}$	$\{S, C\}$	$\{S, A\}$	
$\{B\}$	$\{A, C\}$	$\{A, C\}$	$\{B\}$	$\{A, C\}$
b	a	a	b	a

Final Triangular Table

$\{S, A, C\}$	$\leftarrow X_{1,5}$			
$\emptyset$	$\{S, A, C\}$			
$\emptyset$	$\{B\}$	$\{B\}$		
$\{S, A\}$	$\{B\}$	$\{S, C\}$	$\{S, A\}$	
$\{B\}$	$\{A, C\}$	$\{A, C\}$	$\{B\}$	$\{A, C\}$
b	a	a	b	a

- Table for string 'w' that has length 5
- The algorithm populates the triangular table

Example (Result)

- Is baaba in $L(G)$?

Yes

We can see the S in the set X_{1n} where 'n' = 5

We can see the table

the cell $X_{15} = (S, A, C)$ then

if $S \in X_{15}$ then baaba $\in L(G)$

Theorem

- The CYK Algorithm correctly computes X_{ij} for all i and j ; thus w is in $L(G)$ if and only if S is in X_{1n} .
- The running time of the algorithm is $O(n^3)$.

References

- J. E. Hopcroft, R. Motwani, J. D. Ullman, *Introduction to Automata Theory, Languages and Computation*, Second Edition, Addison Wesley, 2001
- T.A. Sudkamp, *An Introduction to the Theory of Computer Science Languages and Machines*, Third Edition, Addison Wesley, 2006